

Fundamentals of Finance

Course: ATD — Accounting Technicians Diploma

Level: Level III

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Syllabus version: Based on the standard KASNEB ATD Level III Fundamentals of Finance syllabus topics — the finance function, sources of finance, time value of money, and basic investment appraisal. Authored from general finance knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

Original work notice. Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

Contents

1. The finance function in an organisation
 2. Sources of finance — short-term
 3. Sources of finance — long-term
 4. The time value of money
 5. Present value and future value
 6. Working capital management, basics
 7. Investment appraisal — payback period
 8. Investment appraisal — net present value (NPV)
 9. Full worked question — NPV of a small project
 10. Practice examination — KASNEB-style paper (original, not a past paper)
 11. Marking guide — model answers to the practice examination
 12. Exam technique and revision checklist
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1. The finance function in an organisation

The **finance function** is responsible for raising funds needed by the organisation, deciding how to invest those funds (in projects and assets), and managing day-to-day financial resources (working capital, cash) to keep the organisation solvent while pursuing its objectives — commonly summarised as three interrelated decisions: the **financing decision** (how to raise funds), the **investment decision** (which projects/assets to invest in), and the **dividend/distribution decision** (how much profit to distribute to owners versus retain for reinvestment).

2. Sources of finance — short-term

- **Trade credit** — buying on credit from suppliers, effectively financing short-term needs without a separate loan agreement.
 - **Bank overdraft** — a flexible short-term facility allowing a business to withdraw more than its account balance, up to an agreed limit, useful for managing temporary cash-flow gaps.
 - **Short-term bank loans** — borrowed for a defined short period, typically for a specific working-capital need.
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3. Sources of finance — long-term

- **Owner's capital/equity** — funds invested by the business owner(s) or shareholders; no fixed repayment obligation, but owners share in the risk and reward of the business.
 - **Long-term bank loans** — borrowed over a period of several years, typically for financing non-current assets, usually requiring security/collateral.
 - **Debentures/bonds** — long-term debt instruments issued to investors, carrying a fixed interest obligation and a specified repayment date.
 - **Retained earnings** — profit kept within the business rather than distributed to owners, reinvested to fund growth without needing to raise external finance.
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4. The time value of money

The **time value of money** is the principle that a given sum of money is worth more today than the same nominal sum in the future, because money available today can be invested to earn a return, and because inflation typically erodes future purchasing power. This principle underlies almost all finance decision-making, since it means cash flows occurring at different

points in time cannot be validly compared without first adjusting them to a common point in time.

5. Present value and future value

Future value (FV) — what a sum invested today will grow to after a number of periods at a given interest rate:

$$FV = \text{Present value (PV)} \times (1 + r)^n$$

Present value (PV) — what a future sum is worth today, given a specified discount rate (essentially, the reverse calculation):

$$PV = \text{Future value (FV)} \div (1 + r)^n$$

where **r** is the interest/discount rate per period and **n** is the number of periods. **Discounting** — converting a future cash flow into its present value — is central to how investment projects are evaluated.

6. Working capital management, basics

Working capital = Current assets – Current liabilities — the funds available for a business's day-to-day operations. Effective working capital management involves balancing: holding enough inventory to meet customer demand without tying up excessive cash; collecting receivables from customers promptly, without damaging customer relationships through overly aggressive terms; and managing payables to suppliers to make the best use of available credit terms, without damaging supplier relationships or missing early-payment discounts where they're worthwhile. Too little working capital risks the business being unable to meet short-term obligations; too much ties up funds that could otherwise be productively invested elsewhere.

7. Investment appraisal — payback period

The **payback period** is the length of time required for the cumulative cash inflows from an investment project to equal the original investment outlay.

$$\text{Payback period (for even annual cash inflows)} = \text{Initial investment} \div \text{Annual cash inflow}$$

A shorter payback period is generally preferred (all else equal), since it means the investment recovers its cost — and reduces the business's risk exposure — more quickly. The main

limitation of payback period is that it ignores both the time value of money and any cash flows occurring after the payback point.

8. Investment appraisal — net present value (NPV)

Net present value (NPV) discounts all of a project's expected future cash flows to their present value (using an appropriate discount rate reflecting the cost of capital/required return), and subtracts the initial investment:

$$\text{NPV} = (\text{Sum of discounted future cash inflows}) - \text{Initial investment}$$

A project with a **positive NPV** is expected to increase the business's wealth (in present-value terms) and is generally acceptable; a **negative NPV** suggests the project would reduce wealth and should generally be rejected. Unlike payback period, NPV properly accounts for the time value of money and considers all cash flows over the project's life.

9. Full worked question — NPV of a small project

Scenario. A project requires an initial investment of KSh 400,000 today, and is expected to generate cash inflows of KSh 250,000 at the end of year 1 and KSh 220,000 at the end of year 2. The relevant discount rate is 10% per year.

Required: Compute the NPV of the project and advise whether it should be accepted.

Solution:

$$\text{PV of year 1 cash inflow} = 250,000 \div (1.10)^1 = 250,000 \div 1.10 = \text{KSh } 227,273$$

$$\text{PV of year 2 cash inflow} = 220,000 \div (1.10)^2 = 220,000 \div 1.21 = \text{KSh } 181,818$$

$$\text{Total PV of inflows} = 227,273 + 181,818 = \text{KSh } 409,091$$

$$\text{NPV} = 409,091 - 400,000 = \text{KSh } 9,091 \text{ (positive)}$$

Advice: Since the NPV is positive, the project is expected to increase the business's wealth in present-value terms, and should be accepted (assuming the discount rate used correctly reflects the project's risk and the business's cost of capital).

10. Practice examination — KASNEB-style paper (original, not a past paper)

Note on format. This section is laid out the way a KASNEB ATD examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam

conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

ATD LEVEL III — FUNDAMENTALS OF FINANCE (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

- (a) Outline the THREE key decisions typically associated with the finance function. *(9 marks)*
 - (b) Outline THREE short-term sources of finance available to a business. *(6 marks)*
 - (c) Outline TWO long-term sources of finance, with one advantage of each. *(5 marks)*
- (Total: 20 marks)*
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QUESTION TWO

- (a) Explain the time value of money concept and TWO reasons underlying it. *(8 marks)*
 - (b) Compute the future value of KSh 200,000 invested today at 9% per year for 3 years. *(6 marks)*
 - (c) Compute the present value of KSh 300,000 receivable in 2 years' time, at a discount rate of 10% per year. *(6 marks)*
- (Total: 20 marks)*
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QUESTION THREE

- (a) Define "working capital" and explain the risk of holding too little of it. *(8 marks)*
 - (b) Explain the risk of holding too much working capital. *(6 marks)*
 - (c) Outline TWO practical measures a business can take to manage its trade receivables effectively. *(6 marks)*
- (Total: 20 marks)*
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QUESTION FOUR

A project requires an initial investment of KSh 600,000 and generates equal annual cash inflows of KSh 150,000.

Required:

(a) Compute the payback period. (6 marks)

(b) Outline TWO limitations of the payback period as an investment appraisal method. (8 marks)

(c) Distinguish payback period from net present value in terms of how they treat the time value of money. (6 marks)

(Total: 20 marks)

QUESTION FIVE

A project requires an initial investment of KSh 500,000 today and is expected to generate a single cash inflow of KSh 650,000 at the end of year 2. The relevant discount rate is 12% per year.

(a) Compute the present value of the year-2 cash inflow. (8 marks)

(b) Compute the NPV of the project. (6 marks)

(c) Advise, with a reason, whether the project should be accepted. (6 marks)

(Total: 20 marks)

11. Marking guide — model answers to the practice examination

QUESTION ONE

(a) (9 marks — 3 marks each) — Financing decision (how to raise funds); investment decision (which projects/assets to invest in); dividend/distribution decision (how much profit to distribute versus retain).

(b) (6 marks — 2 marks each) — Trade credit, bank overdraft, short-term bank loans.

(c) (5 marks) — Long-term bank loans (funds larger, long-term purchases, e.g. non-current assets) — retained earnings (no need to raise external finance, no dilution of ownership) — any two named sources with one correct advantage each.

QUESTION TWO

(a) (8 marks) — Money available today is worth more than the same sum in future, because it can be invested to earn a return, and because inflation erodes future purchasing power.

(b) (6 marks) — $FV = 200,000 \times (1.09)^3 = 200,000 \times 1.295029 = \text{KSh } 259,006$ (approximately).

(c) (6 marks) — $PV = 300,000 \div (1.10)^2 = 300,000 \div 1.21 = \text{KSh } 247,934$ (approximately).

QUESTION THREE

- (a) (8 marks) — Working capital = Current assets – Current liabilities; too little risks the business being unable to meet its short-term obligations as they fall due, potentially damaging supplier/creditor relationships or even solvency.
- (b) (6 marks) — Too much working capital ties up funds unnecessarily (e.g. excess inventory or cash) that could otherwise be productively invested elsewhere, reducing overall returns to the business.
- (c) (6 marks) — Setting clear, appropriately enforced credit terms; following up overdue accounts promptly (offering early-payment discounts also acceptable) — any two, 3 marks each.

QUESTION FOUR

- (a) (6 marks) — Payback period = $600,000 \div 150,000 = 4$ years.
- (b) (8 marks – 4 marks each) — Ignores the time value of money; ignores cash flows occurring after the payback point.
- (c) (6 marks) — Payback period ignores the time value of money entirely (treats all cash flows as equally valuable regardless of timing); NPV explicitly discounts all cash flows to present value, properly accounting for the time value of money.

QUESTION FIVE

- (a) (8 marks) — $PV = 650,000 \div (1.12)^2 = 650,000 \div 1.2544 = \text{KSh } 518,304$ (approximately).
- (b) (6 marks) — $NPV = 518,304 - 500,000 = \text{KSh } 18,304$ (positive).
- (c) (6 marks) — Since NPV is positive, the project is expected to increase the business's wealth in present-value terms and should be **accepted**, assuming the 12% discount rate correctly reflects the project's risk and the business's cost of capital.

12. Exam technique and revision checklist

- Always state the relevant formula (FV, PV, payback, NPV) explicitly before substituting numbers — computational marks are typically awarded step by step.
- Present/future value questions: be careful to raise $(1+r)$ to the correct power matching the number of periods — a very common source of avoidable errors.
- Payback vs NPV "distinguish" questions: anchor your answer specifically to the time-value-of-money treatment — this is almost always the examiner's intended contrast point.
- NPV questions: always state your final accept/reject advice explicitly, tied to whether NPV is positive or negative — a computation without a clear recommendation may lose marks on an "advise" question.

- Working capital questions: address both the "too little" and "too much" risk explicitly if asked generally about working capital management — a one-sided answer only partially addresses the topic.

Quick revision — one line per topic:

1. Finance function: financing decision, investment decision, dividend decision.
2. Short-term finance: trade credit, overdraft, short-term loans. Long-term finance: equity, long-term loans, debentures, retained earnings.
3. Time value of money: money today is worth more than the same sum in future (can be invested; inflation erodes future value).
4. $FV = PV \times (1+r)^n$; $PV = FV \div (1+r)^n$.
5. Working capital = Current assets – Current liabilities; too little risks insolvency, too much ties up funds unproductively.
6. Payback period = Initial investment $\div$ Annual cash inflow (for even inflows); ignores time value of money and post-payback cash flows.
7. NPV = PV of future cash inflows – Initial investment; positive NPV $\rightarrow$ accept, negative $\rightarrow$ reject.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.